

Support tickets dataset cleaned from 148 to 134 documents, raising quality score from 71.2 to 98.7.

Data cleaning raised the quality score from 71.2 out of 100 to 98.7 out of 100 by removing 14 problematic documents, reducing the dataset from 148 to 134 documents. The cleaned dataset contains 53 billing documents, 47 shipping documents, and 34 technical documents, yielding an imbalance ratio of 1.56. PII was successfully reduced from 18 documents to 0 documents, and 1 document still contains garbled characters.

Data quality before	After cleaning	Documents	Numbers fact-checked
71.2/100	98.7/100	148 → 134	26

What this dataset is: This dataset consists of customer support tickets categorized into billing, shipping, and technical issues based on their folder names, intended for text classification.

Findings

INFO Overall data quality score improved after cleaning

Data cleaning raised the quality score from 71.2 out of 100 to 98.7 out of 100 by removing 14 problematic documents, reducing the dataset from 148 to 134 documents.

Recommendation: Use the cleaned dataset of 134 documents for baseline model training.

Evidence: cleaning_summary_001, txt_quality_score_001, txt_quality_score_002

WARNING Potential label misclassifications detected

Centroid similarity analysis showed 97.12% agreement with assigned categories, identifying 4 documents that read more like another label.

Recommendation: Manually review and re-label the 4 flagged documents prior to model training.

Evidence: txt_labels_001

INFO PII present in raw text fully masked

Raw data contained 18 documents with personal data representing 12.16% of all documents, which was successfully reduced to 0 documents and 0.0% share in the cleaned dataset.

Recommendation: Ensure PII masking scripts are integrated into production data ingestion pipelines.

Evidence: txt_pii_001, txt_pii_002

WARNING Duplicates and cross-label ambiguities present in raw text

Raw corpus analysis revealed 4 exact extra copies, 5 near-duplicate extra copies, and 1 group appearing across multiple labels.

Recommendation: Remove duplicate tickets and eliminate cross-label duplicates to prevent test set contamination.

Evidence: txt_dupes_001

WARNING Widespread email confidentiality boilerplate detected

A single recurring confidentiality disclaimer appeared in 57 documents, representing 38.51% of the entire raw document set.

Recommendation: Strip standardized disclaimers and footers so feature extraction focuses on ticket content.

Evidence: txt_boilerplate_001

INFO Non-English support tickets filtered out

The raw dataset contained 5 documents in non-English languages, whereas after language filtering 0 non-English documents remain and English accounts for 100.0% of documents.

Recommendation: Route non-English tickets to language-specific classifiers or translation preprocessing steps.

Evidence: `txt_language_001`, `txt_language_002`

INFO Moderate label balance maintained across ticket classes

The cleaned dataset contains 53 billing documents, 47 shipping documents, and 34 technical documents, yielding an imbalance ratio of 1.56.

Recommendation: Use stratified split strategies during train-test partitioning to maintain balanced class proportions.

Evidence: `txt_balance_002`

WARNING Encoding damage and garbled characters remain

Raw text analysis identified 3 documents with encoding damage, and after cleaning, 1 document still contains garbled characters.

Recommendation: Manually inspect and repair the remaining document with encoding damage to prevent tokenization issues.

Evidence: `txt_quality_001`, `txt_quality_002`

Next steps

1. Inspect the 1 document containing garbled characters due to remaining encoding damage.
2. Review the 4 documents flagged by centroid similarity as potential label misclassifications.
3. Proceed with training the support ticket classifier on the 134 cleaned documents.

How the agents checked themselves

Model calls 8 · guardrail catches 0 · self-corrections 1 · review rounds 2 · model switches 6

+12s Switched models for the strategist

`gemini-3.8-flash` hit its server error 503 (set aside for 30s); trying the next model in the pool.

+25s Switched models for the analyst

`gemini-3.7-flash` hit its server error 503 (set aside for 30s); trying the next model in the pool.

+51s Switched models for the reviewer

`gemini-3.8-flash` hit its daily limit; trying the next model in the pool.

+69s Reviewer requested 1 change(s) and 1 addition(s)

Finding 3 (severity): Finding 3 is classified as critical even though PII has already been successfully masked in the cleaned dataset. Fix: Change the severity from critical to info or warning because the privacy issue has been resolved. Missed: Encoding damage and garbled characters present in t...

+70s Switched models for the analyst

`gemini-3.7-flash` hit its daily limit; trying the next model in the pool.

+70s Switched models for the analyst

`gemini-3.6-flash` hit its daily limit; trying the next model in the pool.

+110s Switched models for the reviewer

`gemini-3.5-flash` hit its server error 503 (set aside for 30s); trying the next model in the pool.

+113s Reviewer approved the findings

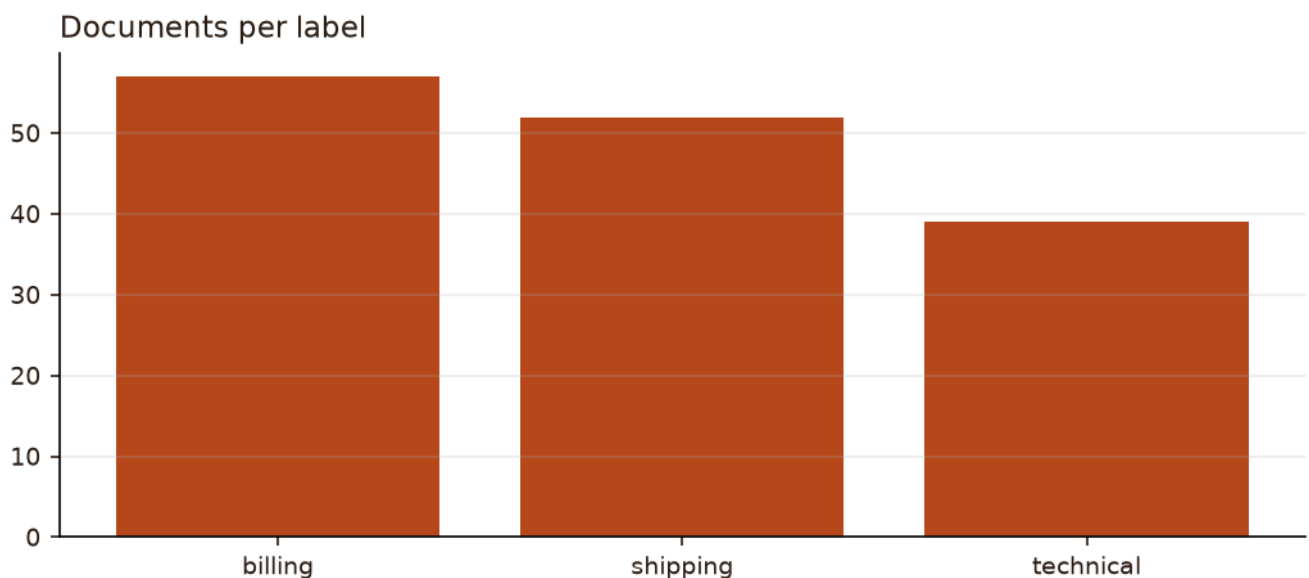
Cleaning applied

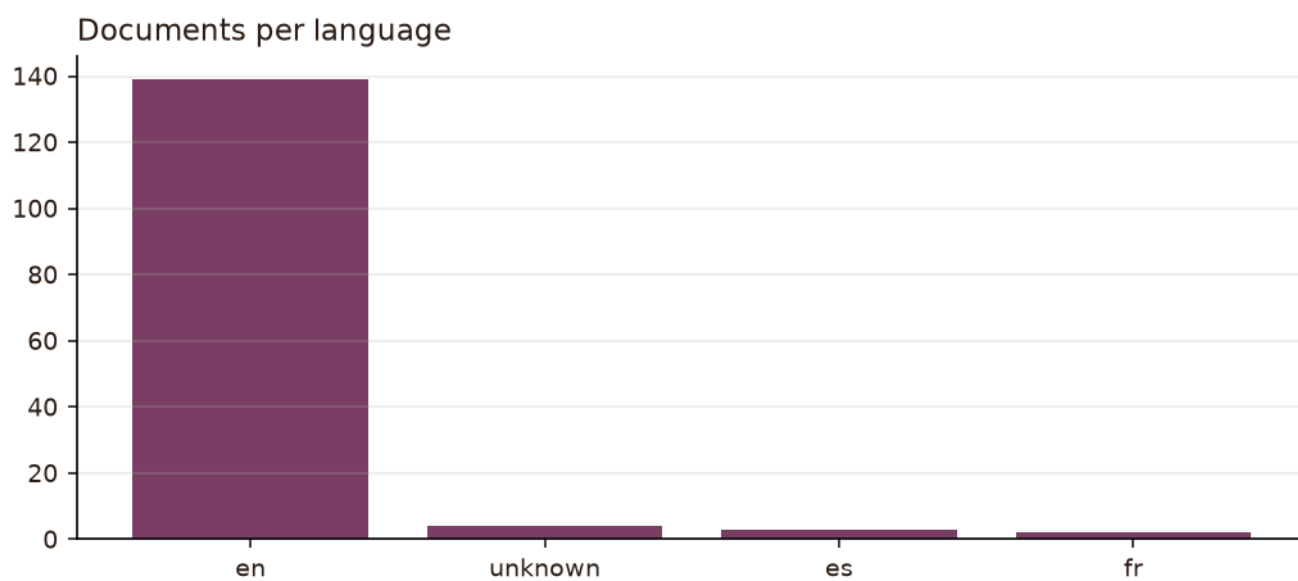
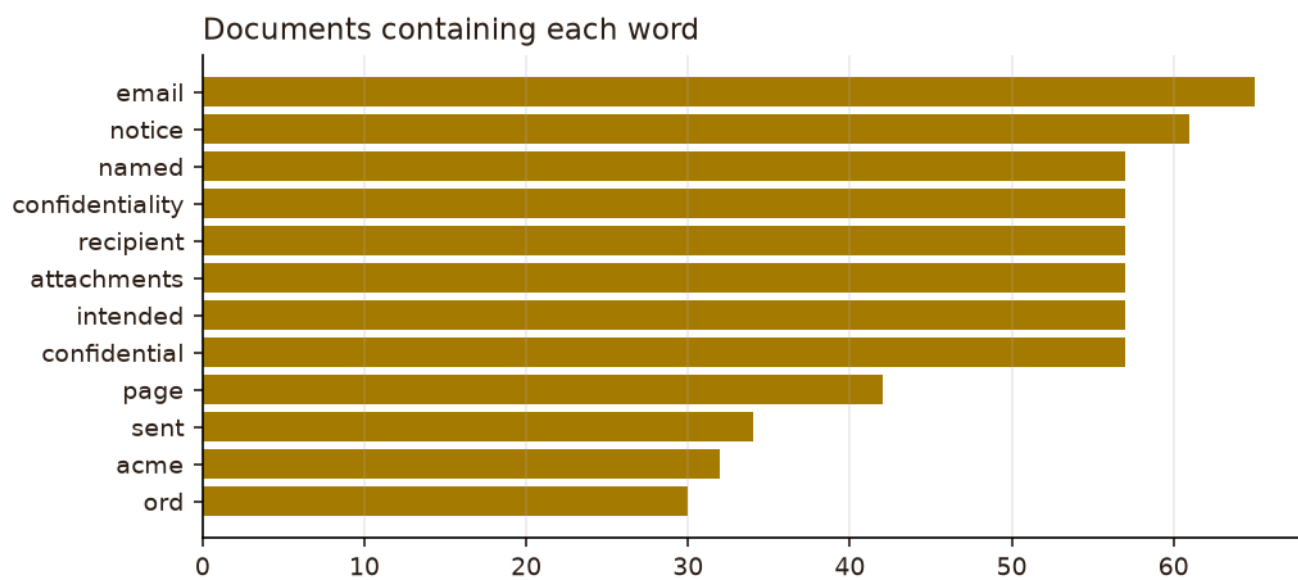
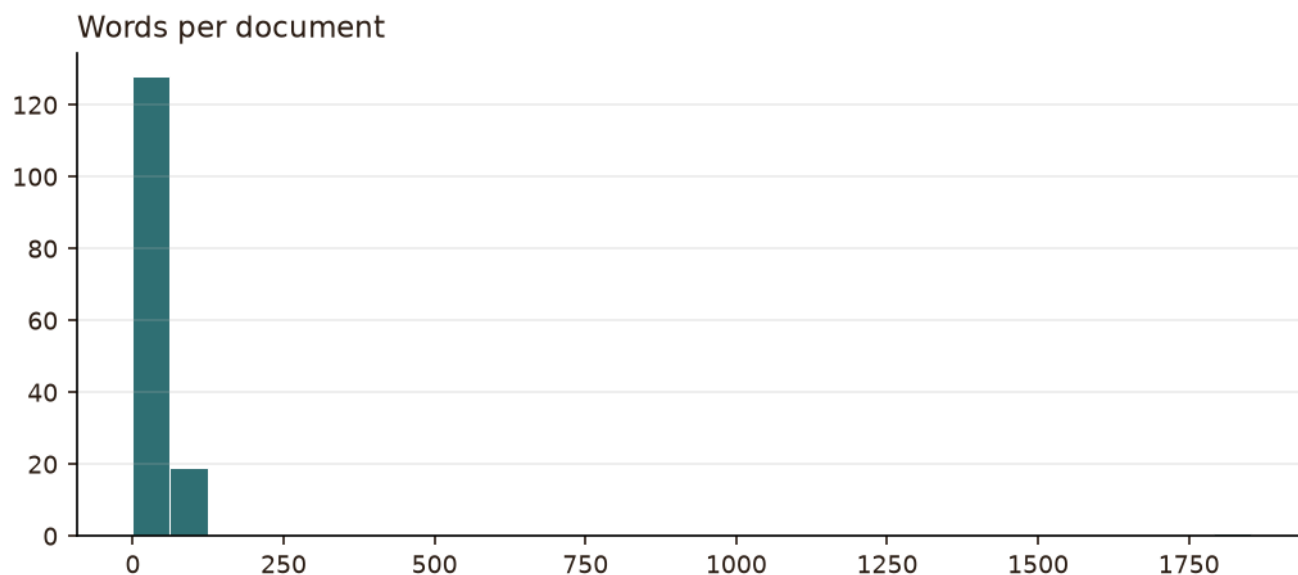
This cleaning plan repairs text encodings, strips HTML markup, normalizes whitespace, removes recurring boilerplate disclaimers, and masks sensitive PII. It then deduplicates documents, filters out non-English messages, removes empty/too-short tickets, and flags suspected label misclassifications for human review.

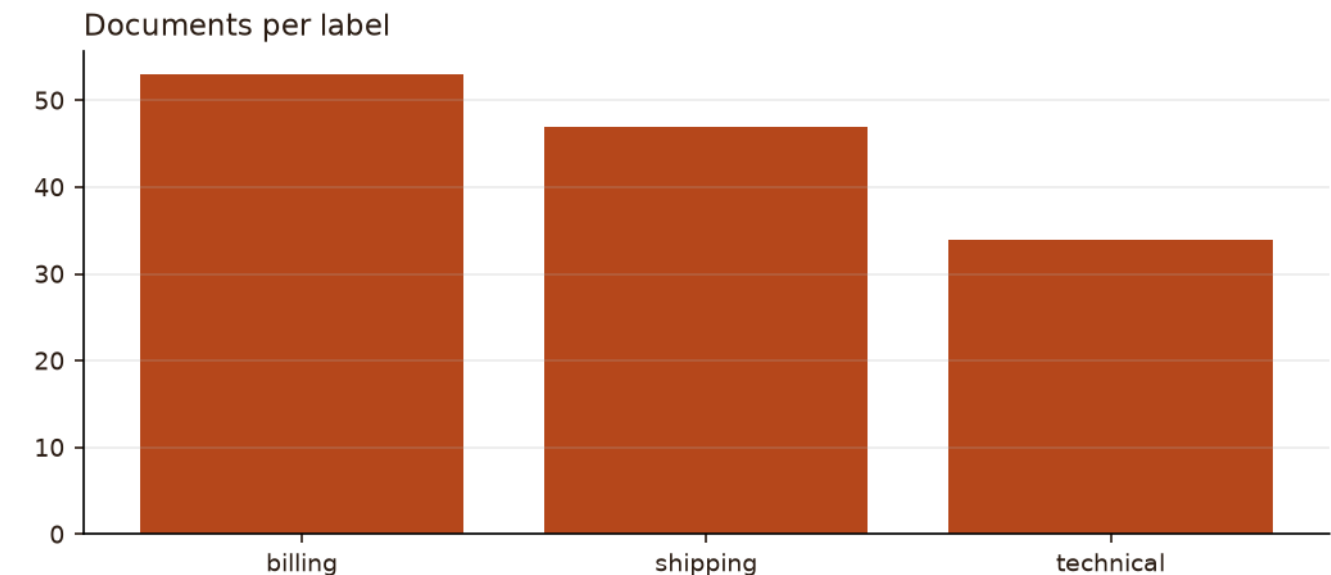
#	Operation	Risk	Columns	Documents	Why
1	fix_encoding	safe	all	148	Repair garbled characters detected across 3 support tickets.
2	strip_html	safe	all	148	Remove HTML tags and entities found in 3 documents.
3	normalize_whitespace	safe	all	148	Collapse irregular spacing and blank lines across cleaned documents.
4	strip_boilerplate	lossy	all	148	Remove recurring confidentiality notice lines present in over 38% of documents.
5	mask_pii	lossy	all	148	Mask email addresses, card numbers, phone numbers, and SSNs detected in 18 documents.
6	drop_exact_duplicates	destructive	all	148 → 143	Remove redundant exact duplicate tickets across the corpus.
7	drop_near_duplicates	destructive	all	143 → 142	Remove near-duplicate tickets sharing high MinHash similarity.
8	drop_cross_label_duplicates	destructive	all	142	Remove ambiguous tickets that appear as near duplicates across different label categories.
9	drop_short_documents	destructive	all	142 → 139	Remove empty and uninformative documents with fewer than 5 words.
10	filter_language	destructive	all	139 → 134	Filter out foreign-language documents (es, fr) while retaining standard English support tickets.
11	flag_suspected_mislabels	safe	all	134	Flag the 4 documents identified by centroid comparison as likely label misclassifications.

The same steps are in `cleaning_pipeline.py`, which reruns on the full data.

Charts







Label balance

Class	Documents	Share %
billing	57	38.5135
shipping	52	35.1351
technical	39	26.3514

Largest class / smallest class: 1.4615

Documents that read like another label (checked by code)

Document	Label	Reads like	Text (masked)
billing/ticket_bill_052.txt	billing	shipping	Hello, My package has not arrived and the tracking number says delivered. The courier left the pa...
technical/ticket_tech_032.txt	technical	billing	Hello team, I was charged twice for my Team Workspace subscription this month. Why is there a lat...
technical/ticket_tech_033.txt	technical	billing	Hello team, I was charged twice for my subscription and need a refund of the extra payment on my...
shipping/ticket_ship_047.txt	shipping	technical	Hi, The app crashes every time I log in after the latest update, and sync stopped working with an...

Topics

Topic	Strongest words	Share %
1	payment, card, email, charge, billing, notice	38.5135
2	delivery, parcel, order, left, tracking, address	33.1081
3	returns, timeout, work, stopped, files, used	25.0

Every number in the findings was checked against the evidence artifacts listed. Models used: gemini-3.5-flash, gemini-3.5-flash-lite, gemini-3.6-flash, gemini-3.7-flash.